

Using Python to Construct a GUI for Plotting Field Lines of Static Charges

For various reasons, physicists often need to plot field lines of static charges or magnetic fields. This can be done beautifully using Python. A Python GUI makes the plotting simple and easy.



In layman's terms, the graphical user interface (GUI) is a series of labels, buttons, text boxes and many other widgets. The conventional way of taking inputs from users is through the shell of a compiler. The GUI gives programming a professional look and takes inputs from the user through a separate window.

This is an article about plotting the field lines of static charges. In books on electrostatics and magnetism, generally a plot of the field lines of a dipole is given, but rarely is anything beyond dipoles looked at. This program will help in visualising the plot of as many charges as possible within the limits of a certain coordinate axis.

We have used the IDLE 3.5.2 IDE (integrated development environment) for this program.

Concepts used

NumPy provides tools such as:

- An array object of arbitrary homogeneous items
- Fast mathematical operations over arrays

- Linear algebra, Fourier transforms, random number generation
- SciPy provides a scientific computing package for Python. It imports all the functions from the NumPy namespace.

Tkinter GUI

A range of user friendly GUIs is available in Python. Some of them are Tkinter, wxPython, PythonWin, PyQt, PyGTK, Java Swing, etc. The most famous graphics module in Python is Tkinter.

Tkinter offers a wide variety of widgets for a graphical user interface, including Frame, Label, Entry, Text, Canvas, Button, Radiobutton, Checkbutton, Scale, Listbox, Scrollbar, OptionMenu, Spinbox, LabelFrame and PanedWindow.

Class

A class is the building block of object oriented programming (OOP) in Python, and is initialised by using the class keyword.